Map of MALDER results

2025-03-19

```
res_dir <- "../results/s07_d_MALDER_map/"
```

Load data

```
# Load MALDER results
dates_full <- readRDS("../results/s07_c_MALDER/malder_results_all.RData")

# Subset to `ref A`$Borneo and `ref B`$Mozambique Chopi
dates <- dates_full %>%
  filter(
    `ref A` == "Borneo" &
      `ref B` == "Mozambique_Chopi"
  )

# Load KML
#kml_data <- sf::st_read(dsn = "../data/raw/data_input/Data_with_pierron.kml")
#kml_data <- sf::st_transform(kml_data, crs = 29702)
# Add names
#kml_data$Name <- c("Merina", "Sakalava_Tsimihety", "Betsimisaraka",
#                   "Betsileo", "Diego", "Temoro", "Vezo", "Mikea",
#                   "Mikea", "Mikea", "Mikea", "Mikea", "Mikea", "Mikea")


kml_data_full <- readRDS("../data/raw/data_input/sampling_locations_with_pierron.rds")
```

To create a map showing estimated admixture timing for both Merina A and Merina B, we create a point for Merina B by adjusting the x coordinate to be slightly south of Merina A. See below.

```
group <- c("MerinaA", "MerinaB")
x <- c(519090.2, 519090.2)
y <- c(834836.2, 810000.0)
sample <- c(0,0)

merina_coords <- data.frame(Name = group, x = x, y = y, sample = sample)

merina_coords <- merina_coords %>%
  st_as_sf(coords = c("x", "y"), crs = 29702) %>%
  mutate(
    x = st_coordinates(.)[, 1],
    y = st_coordinates(.)[, 2]
  )
```

```r
kml_data <- kml_data_full %>%
  mutate(
    x = st_coordinates(.)[, 1],
    y = st_coordinates(.)[, 2]
  )

# rbind the two datasets
kml_data_all <- rbind(kml_data, merina_coords)
# remove Merina
kml_data <- kml_data_all %>%
  filter(Name != "Merina")
```

```r
# match kml_data$Name and dates rownames
matching <- match(kml_data$Name, as.character(dates$test_pop))

# add admixture date and std_err to kml_data
kml_data$date <- dates[matching, "admixture_date"]
kml_data$std_err <- dates[matching, "CI"]

kml_data$sample <- ifelse(kml_data$Name %in% c("Merina","MerinaA", "MerinaB", "Sakalava_Tsimihety", "Be

kml_data <- kml_data[order(kml_data$Name), ]

# Add plot shape
kml_data$shape <- c(1, 3, 7, 15, 8, 17, 17, 17, 17, 17, 17, 17, 19, 21, 23)

# add admixture date in years (gen time 30 years)
kml_data$years <- paste(
  round(kml_data$date * 30, 0),
  "\u00b1",
  round(kml_data$std_err * 30, 0), "BP"
)


labels_data <- kml_data %>%
  group_by(Name) %>%
  reframe(
    x = mean(x),
    y = mean(y),
    date = years,
    geometry = st_centroid(st_union(geometry))
  )

# Mikea has multiple locations, remove duplicates and leave one row to label
labels_data <- labels_data[!duplicated(labels_data$Name), ]
```

Create map

```r
# List possibly relevant countries
target <- c("Madagascar", "Mozambique", "Comoros", "Mauritius", "Seychelles", "Mayotte", "United Republi
```

```r
#Create an inset of Madagascar
madagascar_inset <- ne_countries(
  country = target,
  type = "map_units",
  scale = 50
)

madagascar_dissolved <- madagascar_inset %>%
  st_union() %>%
  st_as_sf()
```

```r
# ggplot() +
#   geom_sf(data = madagascar_dissolved) +
#   geom_sf(
#     data = sf::st_as_sf(kml_data),
#     aes(shape = Name, color = Name, fill = Name),
#     size = 2
#   ) +
#   coord_sf(
#     crs = st_crs(29702),
#     xlim = c(-1000000, 1800000),
#     ylim = c(1, 2000000), expand = FALSE
#   ) + # Laborde projection
#   theme(panel.background = element_rect(fill = "#F0F8FF", color = NA)) +
#   scale_shape_manual(values = c(1, 3, 7, 15, 8, 17, 19, 21, 23)) +
#   labs(shape = NULL) +
#   labs(
#     x = NULL, y = NULL,
#     color = "Population",
#     shape = "Population",
#     fill = "Population"
#   ) +
#   geom_label_repel(
#     data = labels_data, force = 4, force_pull = 1,
#     aes(x = x, y = y, label = date), size = 3,
#     nudge_y = 2, segment.color = "transparent"
#   ) +
#   scale_color_brewer(palette = "Dark2") +
#   scale_fill_brewer(palette = "Dark2") +
#   guides()
# ggsave(paste0(res_dir, "malder_dates_st_err.jpg"),
#   width = 7, height = 5, dpi = 300
# )
```

```r
library(data.table)
```

```
##
## Attaching package: 'data.table'
```

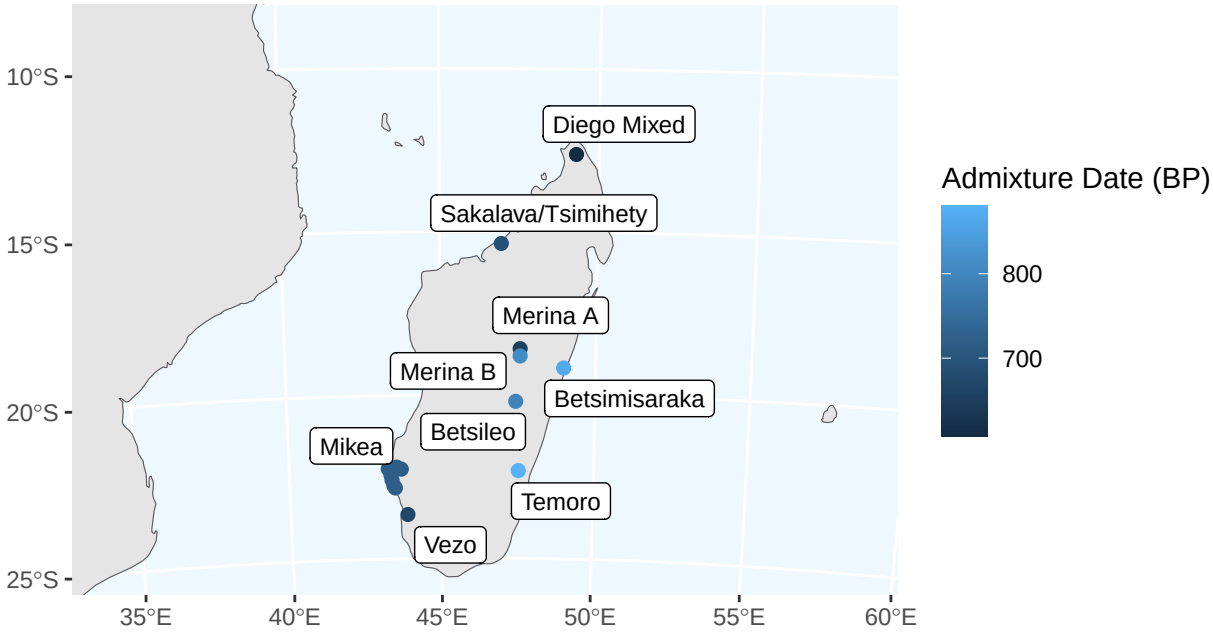
```
## The following objects are masked from 'package:dplyr':
##
##     between, first, last
```

```r
kml_data$yrs <- as.numeric(unlist(tstrsplit(kml_data$years, " ", keep = 1)))

# Replace kml_data$Name == "Sakalava_Tsimihety" with "Sakalava/Tsimihety"
kml_data$Name[kml_data$Name == "Sakalava_Tsimihety"] <- "Sakalava/Tsimihety"
# Replace kmldata$Name == "Diego" with "Diego Mixed"
kml_data$Name[kml_data$Name == "Diego"] <- "Diego Mixed"
# Replace kml_data$Name == "MerinaA" with "Merina A"
kml_data$Name[kml_data$Name == "MerinaA"] <- "Merina A"
# Replace kml_data$Name == "MerinaB" with "Merina B"
kml_data$Name[kml_data$Name == "MerinaB"] <- "Merina B"

labels_data <- kml_data %>%
  group_by(Name) %>%
  summarise(
    x = mean(x),
    y = mean(y),
    geometry = st_centroid(st_union(geometry)) # Optional: Keeps geometry if needed
  )

ggplot() +
  geom_sf(data = madagascar_dissolved) +
  geom_sf(
    data = sf::st_as_sf(kml_data),
    aes(color = yrs, fill = yrs),
    size = 2
  ) +
  coord_sf(
    crs = st_crs(29702),
    xlim = c(-1000000, 1800000),
    ylim = c(1, 2000000),
    expand = FALSE
  ) + # Laborde projection
  theme(panel.background = element_rect(fill = "#F0F8FF", color = NA)#,
        #legend.position = c(0.85, 0.56),
    #legend.background = element_rect(fill = "white")
  ) +
  labs(
    colour = "Admixture Date (BP)",
    fill = "Admixture Date (BP)",
    x = NULL, y = NULL
  ) +
  geom_label_repel(data = labels_data, aes(x = x, y = y, label = Name), size = 3, segment.color = 'trans
  guides()
```




```
ggsave(paste0(res_dir, "malder_dates_gradient.jpg"),
  width = 7, height = 4, dpi = 300
)
```

Recording R environment

Part of ensuring our workflow is reproducible is to keep track of the R environment. We will automatically save our R environment into a file called `r_environment.json`.